LAB NO: 2

Objectives

To learn to use MATLAB to (1) generate polynomials, (2) manipulate polynomials, (3) generate transfer functions, (4) manipulate transfer functions, and (5) perform partial-fraction expansions.

Practice Questions

TASK1: Students who are using MATLAB should now **run ch2apB1 through ch2apB8** in Appendix B. This is your first MATLAB exercise. You will learn how to use MATLAB to (1) represent polynomials, (2) find roots of polynomials, (3) multiply polynomials, and (4) find partial-fraction expansions. Finally, Example 2.3 will be solved using MATLAB

1- Use the following MATLAB and Control System Toolbox statement to form the linear, time invariant (LTI) transfer function of Eq. (2.22).

```
F=zpk([],[-1 -2 -2],2)
```

2- Use the following MATLAB statements to help you get Eq. (2.26).

```
numf=2;
denf=poly([-1 -2 -2]);
[K,p,k]=residue...
(numf,denf)
```

3- Use the following MATLAB and Control System Toolbox statement to form the LTI transfer function of Eq. (2.30).

```
F=tf([3],[1 2 5 0])
```

4- Use the following MATLAB and Symbolic Math Toolbox statements to get Eq. (2.38) from Eq. (2.30).

```
syms s
f=ilaplace...
(3/(s*(s^2+2*s+5)));
pretty(f)
```

Control System- CP-402

TASK2: Students who are performing the MATLAB exercises and want to explore the added capability of MATLAB's Symbolic Math Toolbox should now run ch2apF1 and ch2apF2 in Appendix F. You will learn how to construct symbolic objects and then find the inverse Laplace and Laplace transforms of frequency and time functions, respectively.

5- Use the following MATLAB statements to help you get Eq. (2.47).

```
numf=3
denf=[1250]
[K,p,k]=residue(numf, denf)
```

TASK3: Students who are using MATLAB should now run ch2apB9 through ch2apB12 in Appendix B. You will learn how to use MATLAB to create transfer functions with numerators and denominators in polynomial or factored form. You will also learn how to convert between polynomial and factored forms. Finally, you will learn how to use MATLAB to plot time functions.

TASK4: Students who are performing the MATLAB exercises and want to explore the added capability of MATLAB's Symbolic Math Toolbox should now run ch2apF3 in Appendix F
You will learn how to use the Symbolic Math Toolbox to simplify the input of complicated transfer functions as well as improve readability.

6- Use the following MATLAB and Symbolic Math Toolbox statements to help you get Eq. (2.60).

```
syms s
C=1/(s*(s + 2))
C=ilaplace(C)
```

7- Use the following MATLAB statements to plot Eq. (2.60) for t from 0 to 1 sat intervals of 0.01s.

```
t=0:0.01:3; plot...
(t,(1/2 -1/2*exp (-2*t)))
```

Exercise question 1:

(a) $G(s) = \frac{(s^2+3s+10)(s+5)}{(s+3)(s+4)(s^2+2s+100)}$

(b) $G(s) = \frac{s^3+4s^2+2s+6}{(s+8)(s^2+8s+3)(s^2+5s+7)}$

Exercise question 2:

$$F(s) = \frac{10^4(s+5)(s+70)}{s(s+45)(s+55)(s^2+7s+110)(s^2+6s+95)}$$

write a script to find the partial fraction expansion, using MATLAB's **residue** function, and the inverse Laplace transform, using MATLAB's **ilaplace** function.

The MATLAB script must do the following:

1) Calculate the residues of the partial fraction expansion. The residues' values must be stored in the variable **K**

- The **residue** function's input parameters are arrays containing the transfer function's numerator and denominator polynomial coefficients.

2) Determine the roots of the transfer function's denominator. The roots must be stored in the variable **p**

3) Determine the inverse Laplace transform of $F(s)$. The inverse must be stored as a symbolic expression in the variable **invF**

- The **ilaplace** function's input parameter is a symbolic respresentation of the transfer function.

Exercise Question 3: Working with LABView

Objective

To verify the behavior of closed-loop systems as described in Chapter 1 Case Study.

Minimum required software packages

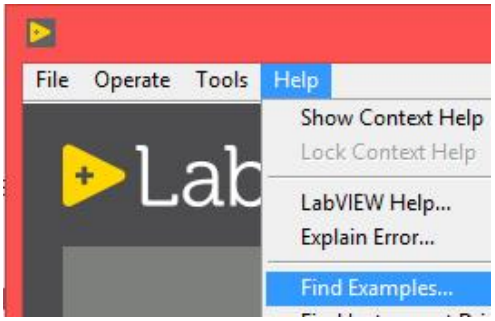
LabVIEW and the LabVIEW Control Design and Simulation Module. *Note*: While no knowledge of LabVIEW is required for this experiment, see the appendix to learn more about LabVIEW, which will be pursued in more detail in later chapters.

Prelab

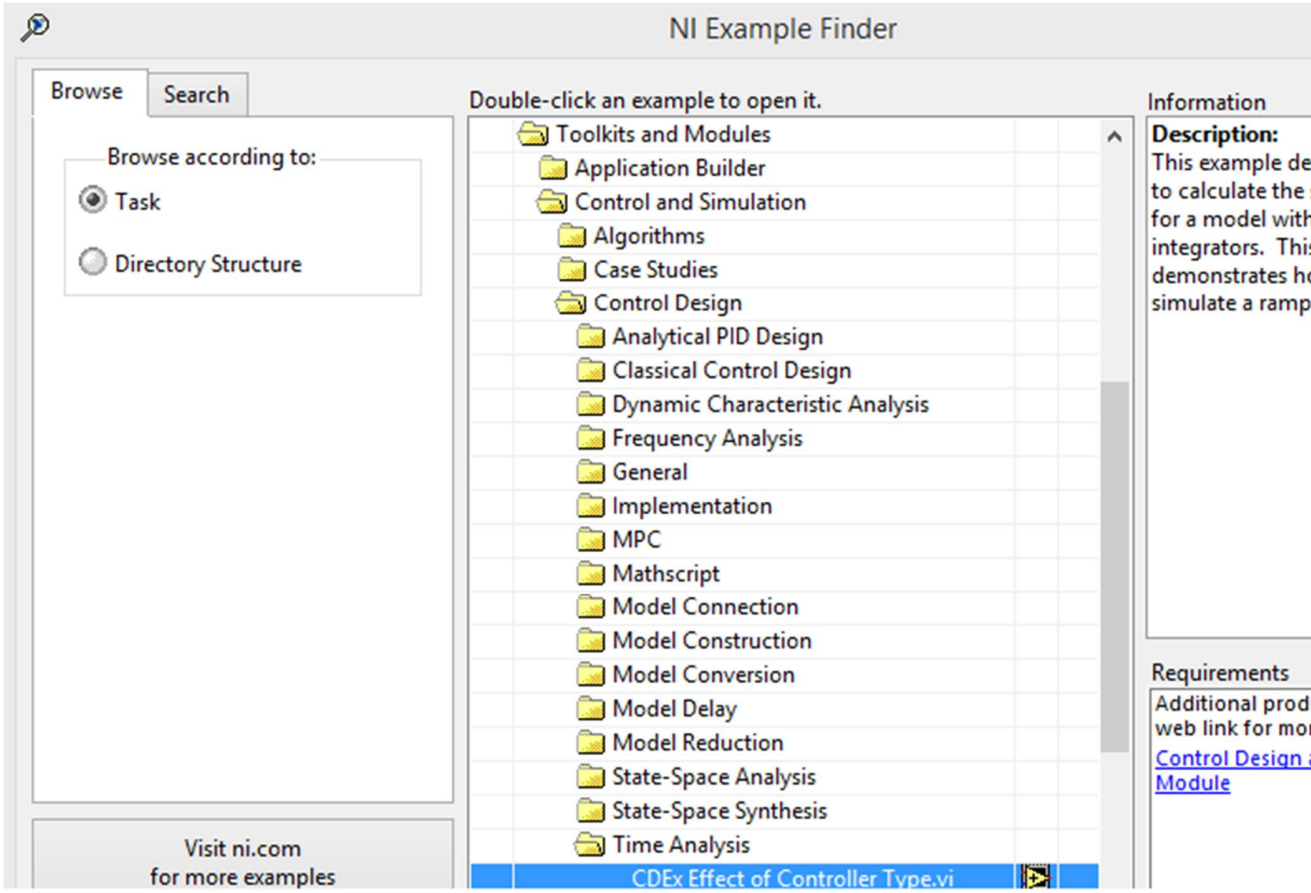
1. Describe the effect of the gain of a closed-loop system upon transient response.

Lab

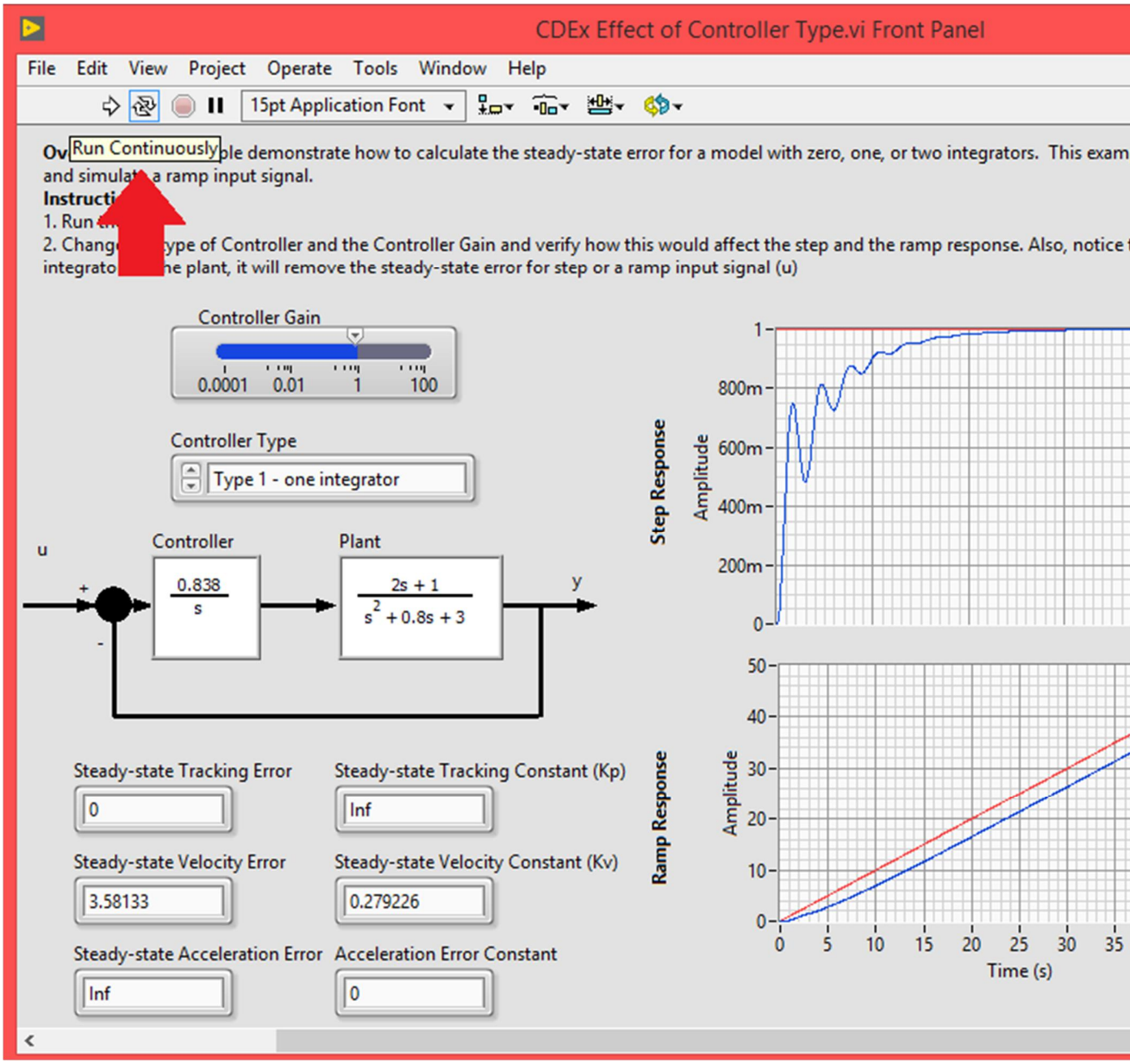
1. Launch LabVIEW and open **Find Examples...** under the **Help** tab.



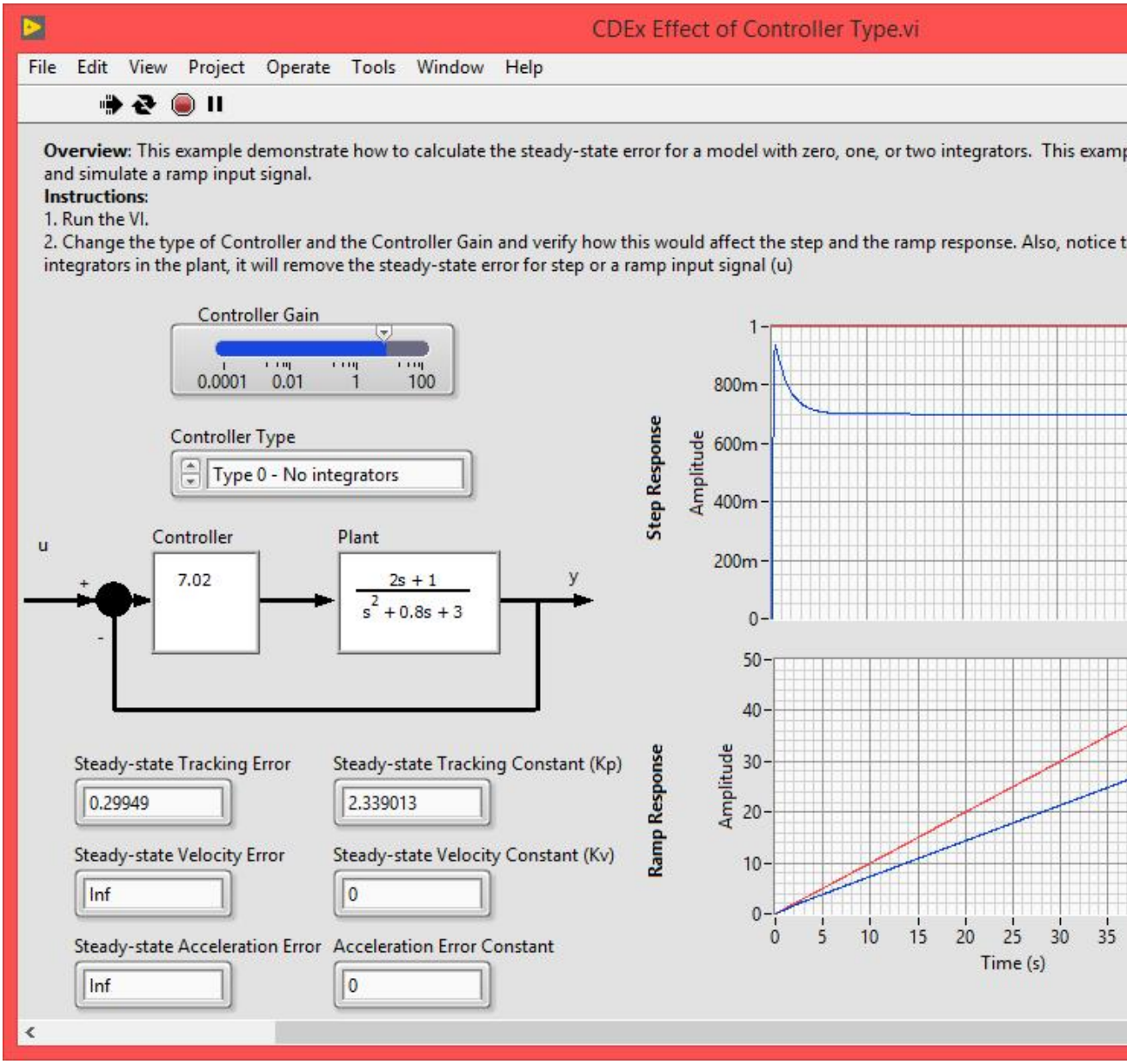
2. In the **NI Example Finder** window, open **CDEx Effect of Controller Type.vi,** found by navigating to it through **Toolkits and Modules/Control and Simulation/Control Design/Time Analysis/CDEx Effect of Controller Type vi.**



3. On the tool bar click the circulating arrows located next to the solid arrow on the left ("Run Continuously"). The program is now running.



4. Move the slider **Controller Gain** and note the effect of high and low gains.
5. Change the controller by clicking the arrows for **Controller Type** and repeat Step 4. Shown below is Type 0 - No integrators.



Postlab

1. Correlate the responses found in the experiment with those described in your Prelab. Explore other examples provided in the LabVIEW example folders.